AI-Driven Traffic Governance for Dhaka

Transforming congestion into smart mobility — an actionable platform that detects street-level violations, simulates outcomes, and enables targeted enforcement for faster, safer, scalable city streets.

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Dhaka Traffic Crisis: The Scale

Economic Loss

$11B lost annually — measurable drag on growth and municipal budgets.

Time Wasted

3.2M hours lost daily — commuters, freight, and emergency responders delayed.

Network Speed

Average travel speeds: 4.8–7 km/h — inefficient and unpredictable mobility.

Root cause: uncontrolled street behavior — not only vehicle volume but micro-level violations that fragment road capacity.

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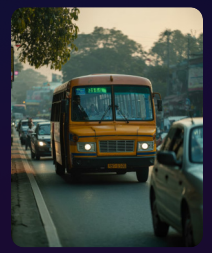
Micro-Level Violations Driving Congestion

- Illegal roadside parking constricting lanes
- Footpath encroachment by vendors forcing pedestrians onto roads
- Random bus stopping disrupting flow and blocking through lanes
- Lane indiscipline and ad-hoc overtaking creating shockwaves



Random Bus Stopping

Creates localized stoppages and queue spillback across intersections.

Footpath Encroachment

Shifts pedestrian flows into vehicular lanes, increasing conflict points.

Impact estimate: these behaviors can reduce usable road capacity by up to 50%, producing dynamic bottlenecks that cascade across the network.

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Beyond Congestion: Public Health Risk

Ambulance Delays

Delays of up to 102 minutes documented — time-critical emergencies suffer.

Survival Impact

Survival falls ~6% per minute of delay for critical cases—traffic equals life-or-death.

Low Ambulance Use

Only ~11% of patients use formal ambulance services, amplifying system fragility.

Conclusion: Traffic mismanagement directly undermines emergency response and public health outcomes.

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Solution Overview

AI-Powered Traffic Governance System — an integrated platform combining real-time detection, predictive simulation, and enforcement workflows to convert chaotic street behavior into data-driven control.

Detect

Computer vision identifies violations: illegal parking, vendor encroachment, random bus stops, lane breaches.

Predict

Digital twin simulations forecast congestion impact and test interventions before implementation.

Act

Automated enforcement alerts, optimized traffic management, and emergency corridor prioritization.

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How It Works: Technology Stack

1

1. Computer Vision (Faster R-CNN)

High-accuracy detection of vehicles, vendors, and violation types at street level.

2

2. Edge AI

Low-latency inference at intersections to enable real-time alerts and local control loops.

3

3. Digital Twin Simulation

“What-if” scenario testing for interventions — re-routing, curb-recall, targeted enforcement.

4

4. Operational Dashboard

Real-time insights, hotspot maps, and enforcement queues for traffic police and city teams.

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Key Features & Outputs

Lane Occupancy & Throughput

Continuous measurement of usable lane capacity and vehicle throughput by segment.

Illegal Parking Detection

Automated alerts with timestamped evidence for enforcement and fine workflows.

Bus Stop Behavior Tracking

Detects unauthorized stopping patterns and quantifies delay impact on trunk routes.

Emergency Corridor Risk Mapping

Identifies high-risk segments and recommends dynamic priority routing for ambulances.

Traffic Improvement Simulation

Test interventions virtually—policy changes, enforcement focus, or infrastructure tweaks—before field rollout.

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Market & Stakeholders

B2G (Primary Buyers)

- Traffic Police — automation & evidence-driven citations
- City Corporations — real-time network monitoring
- Emergency Services — priority routing and corridor management

B2B (Secondary Buyers)

- Logistics & Ride-hailing (Pathao, Truck Lagbe) — route reliability
- Insurance — risk-based premiums using behavioral data
- Supply Chain — predictable lead times and lower dwell

Revenue levers: subscription dashboards, enforcement-as-a-service, prioritized routing licenses, and analytics for private fleets.

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Impact & Value

Economic Recovery

Potential to recover up to 7% of GDP-equivalent losses by restoring flow and productivity.

Delay Reduction

Field trials suggest up to 70% reduction in localized delays when violations are addressed.

Public Safety

Improved emergency response times and higher survival probabilities during critical incidents.

Environmental Gains

Lower idle times reduce emissions by 25–30% on targeted corridors.

Vision: smarter roads enable a smarter economy and safer city life.

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Vision & Next Steps

1

Phase 1 — Pilot

Deploy at 10 high-impact intersections, measure baseline and intervention effects (3–6 months).

2

Phase 2 — Scale

City-wide rollout across priority corridors with integrated enforcement and emergency routing (6–18 months).

3

Phase 3 — Institutionalize

Integrate into Smart Bangladesh 2041 roadmap: policy adoption, funding models, and cross-city replication.

Final statement: “Traffic is not an infrastructure problem — it is a management problem. Management can be automated. Let data guide enforcement, and Dhaka will move.”

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